Lecture L11

*Blind review — please score this lecture on row **L11** in the Excel form (1–10 per dimension).*

Lecture: Backpropagation in Neural Networks

Introduction

Backpropagation is a widely used algorithm for training feedforward neural networks. It allows the information from the cost function to flow backwards through the network, enabling the computation of the gradient. This process is crucial for adjusting the network's parameters, such as weights and biases, in order to minimize the cost and improve the network's performance.

Prerequisites

1. **Feedforward Neural Networks**: A feedforward neural network is a type of artificial neural network where information flows only in one direction, from the input layer to the output layer.
2. **Cost Function**: A cost function, also known as a loss function, measures the difference between the predicted output and the true output. In the context of neural networks, the cost function is usually the mean squared error (MSE) or mean absolute error (MAE).
3. **Gradient Descent**: Gradient descent is an optimization algorithm used to minimize the cost function by adjusting the network's parameters. It does so by taking the gradient of the cost function with respect to the parameters and moving in the direction of the negative gradient.

Core Mechanism/Rules

Forward Propagation

1. **Input Layer**: The input layer receives the input data and forwards it to the hidden layer(s).
2. **Hidden Layer(s)**: The hidden layer(s) processes the input data and produces the output. The output of each hidden layer is the input to the next layer.
3. **Output Layer**: The output layer produces the final output, which is the predicted value.

Backward Propagation

1. **Cost Calculation**: The cost function is computed using the predicted output and the true output.
2. **Gradient Calculation**: The gradient of the cost function with respect to the parameters is calculated using the backpropagation algorithm.

Gradient Calculation

1. **Error Signal**: The error signal, also known as the error vector, is computed as the difference between the true output and the predicted output.
2. **Derivative Calculation**: The derivative of the cost function with respect to the parameters is calculated using the error signal.

Gradient Update

1. **Gradient Adjustment**: The gradient is multiplied by the negative of the learning rate to adjust the network's parameters.

2. **Weight and Bias Updates**: The weights and biases are adjusted based on the gradient and the learning rate.
3. **Gradient Backpropagation**: The gradient is passed backwards through the network, from the output layer to the input layer, to update the weights and biases.
4. **Learning Rate**: The learning rate determines the step size for adjusting the network's parameters. A smaller learning rate allows for more precise adjustments, while a larger learning rate allows for faster convergence.

Variants of Backpropagation

1. **Stochastic Gradient Descent (SGD)**: SGD is a variant of backpropagation that randomly selects a subset of the training data to update the network's parameters at each iteration.
2. **Momentum**: Momentum is a variant of backpropagation that incorporates a momentum term to help the network escape local minima.
3. **AdaGrad**: AdaGrad is a variant of backpropagation that adapts the learning rate for each parameter based on the gradient history.

Limitations

1. **Overfitting**: Backpropagation can lead to overfitting, where the network performs well on the training data but poorly on the test data.
2. **Local Minima**: Backpropagation can get stuck in local minima, where the network's parameters are not optimal but are good enough to fit the training data.
3. **Learning Rate**: Backpropagation requires a good choice of learning rate, which can be difficult to determine.
4. **Initialization**: Backpropagation requires the network's initial weights and biases to be well-chosen.
5. **Complexity**: Backpropagation can be computationally expensive, especially for large neural networks.

By discussing these variants and limitations of backpropagation, we ensure that students are aware of the potential issues and can make informed decisions when applying this technique in their own work.